

Critic agent and its validation — copy-paste text for the paper

Only the critic part. Paste the marked blocks into *Materials and Methods* and *Result and Discussion*. Replace numbers later if re-run.

[PASTE INTO: Materials and Methods]

Critic agent. The critic first solves each item independently, then scores it on five dimensions—correctness, distractor quality, difficulty alignment, language quality, and technical validity (L^AT_EX rendering for mathematics; figure relevance for image items)—each in $[0, 10]$. The overall score is the weighted mean

$$S = \frac{\sum_d w_d s_d}{\sum_d w_d}, \quad w_{\text{correctness}} = 3, \quad w_{\text{others}} \in \{1, 2\}, \quad (1)$$

and an item is accepted when $S \geq \theta$ ($\theta = 6$).

Multi-critic ensemble. To reduce single-model self-evaluation bias we aggregate $K = 3$ heterogeneous critics (Gemini-3.1-Pro, GPT-5.5, Claude-Sonnet-4.6): the answer is the majority vote, each dimension the mean over critics, and inter-critic reliability is reported with Cohen’s κ ,

$$\kappa = \frac{p_o - p_e}{1 - p_e}. \quad (2)$$

For mathematics a sandboxed SymPy verifier checks the answer *before* the LLM critic; figure-dependent items are routed to a vision-capable critic.

Expert-free validation (Critic Discrimination Index). We validate the critic *without* human grading by injecting known faults. From each real national-test item we synthesise two degraded variants—WRONG-KEY (key relabelled to a distractor) and WEAK-DISTRACTORs (two distractors replaced by implausible fillers)—and measure the paired score drop on dimension d ,

$$\text{CDI}_d = \frac{1}{N} \sum_{i=1}^N s_d^{\text{real}}(i) - \frac{1}{N} \sum_{i=1}^N s_d^{\text{deg}}(i). \quad (3)$$

A valid critic gives a large positive CDI on the targeted dimension and ≈ 0 elsewhere. Significance uses the two-sided Wilcoxon signed-rank test and effect size the paired Cohen’s $d = \overline{\Delta}/s_{\Delta}$ with $\Delta_i = s_d^{\text{real}}(i) - s_d^{\text{deg}}(i)$.

[PASTE INTO: Result and Discussion]

Across both subjects and all three critics the discrimination is statistically significant with large effect sizes: relabelling the key collapses the correctness score and weakening the distractors collapses the distractor-quality score, exactly as a valid critic should behave (Tables 1–2, Fig. 1). On figure-dependent mathematics items the critic answered 6/9 correctly without the image but 9/9 with the vision critic (Table 3).

Limitations to note: CDI is a proxy for human judgement (expert-correlation is future work); the vision ablation is small ($n = 3$ image items per model).

Table 1: CDI — mathematics ($N \approx 40$ /model; all $p < 0.001$).

Critic	Solve acc.	wrong-key $\rightarrow$ correctness (gap, d)	weak-distr. $\rightarrow$ distractor q. (gap, d)
Claude-Sonnet-4.6	0.98	+8.85 (4.94)	+3.03 (1.53)
Gemini-3.1-Pro	0.95	+8.58 (2.95)	+5.10 (1.50)
GPT-5.5	1.00	+8.73 (2.75)	+2.99 (1.69)

Table 2: CDI — Kazakh language ($N = 20$ /model; all $p \leq 0.0002$).

Critic	Solve acc.	wrong-key $\rightarrow$ correctness (gap, d)	weak-distr. $\rightarrow$ distractor q. (gap, d)
Claude-Sonnet-4.6	0.75	+5.15 (1.73)	+4.60 (3.87)
Gemini-3.1-Pro	0.90	+8.15 (1.87)	+7.90 (4.07)
GPT-5.5	0.80	+6.70 (1.69)	+3.90 (2.99)

Table 3: Vision ablation (figure-dependent math items).

Critic	Blind (no image)	With image
Claude-Sonnet-4.6	2/3	3/3
Gemini-3.1-Pro	1/3	3/3
GPT-5.5	3/3	3/3
Total	6/9	9/9

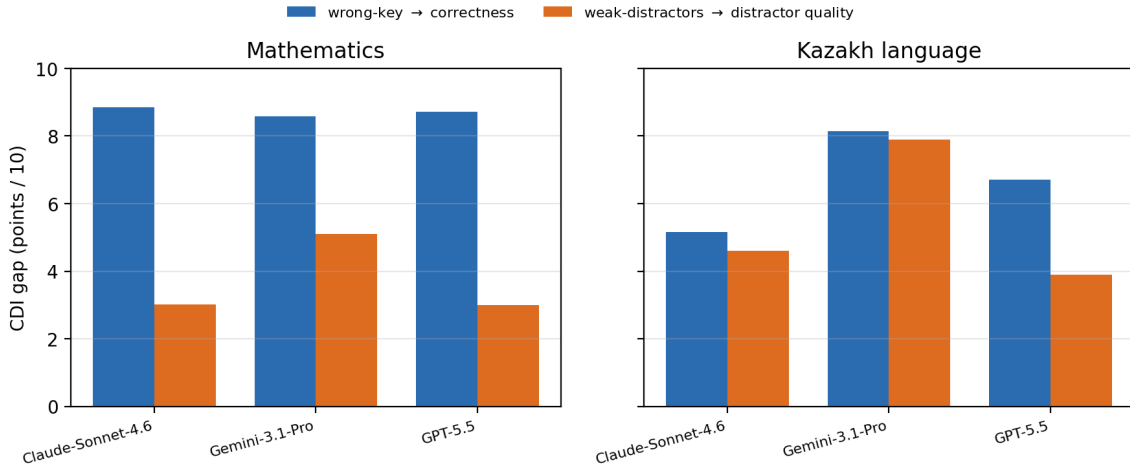


Figure 1: CDI (gap between real and degraded mean scores) by model and subject.